Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For the separation of two miscible liquids which method is used?

(A) Chromatography

(B) Fractional column

(C) Fractional distillation

(D) Separating funnel

Q.2

In Kjeldahl 's method, $CuSO_4$ acts as

(A) oxidising agent

(B) reducing agent

(C) hydrolysing agent

(D) catalytic agent

Q.3

In Lassaigne's test, the organic compound is fused with a piece of sodium metal in order to

(A) increase the ionisation of the compound.

(B) decrease the melting point of the compound.

(C) increase the reactivity of the compound.

(D) convert the covalent compound into a mixture of ionic compounds.

Q.4

A is a lighter phenol and B is an aromatic carboxylic acid. Separation of mixture of A and B can be carried out easily by using a solution of

(A) sodium hydroxide

(B) sodium sulphate

(C) calcium chloride

(D) sodium bicarbonate

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Q.5

Copper wire test for halogens is known as

(A) Duma's Test

(B) Beilstein's Test

(C) Liebig's Test

(D) Lassigne's Test

Q.6

The Lassaigne's extract is boiled with dil. HNO_3 before testing for halogens because

(A) silver halides are soluble in HNO_3

(B) Na_2S and $NaCN$ are decomposed by HNO_3

(C) Ag_2S is soluble in HNO_3

(D) $AgCN$ is soluble in HNO_3

Q.7

The best method for the separation of naphthalene and benzoic acid from their mixture is:

(A) Distillation

(B) Sublimation

(C) Chromatography

(D) Crystallisation

Q.8

Aniline is usually purified by

(A) chromatographic technique

(B) steam distillation

(C) by addition of oxalic acid

(D) fractional crystallization

Q.9

An organic compound has C and H percentage in the ratio $6:1$ and C and O percentage in the ratio $3:4$ the compound is

(A) $HCHO$

(B) CH_3OH

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(C) CH_3CH_2OH

(D) $(COOH)_2$

Q.10

2.79 g of an organic compound when heated in Carius tube with conc. HNO_3 and H_3PO_4 formed is converted into $MgNH_4 \cdot PO_4$ ppt. The ppt. on heating gave 1.332 g of $Mg_2P_2O_7$. The percentage of P in the compound is

(A) 23.33%

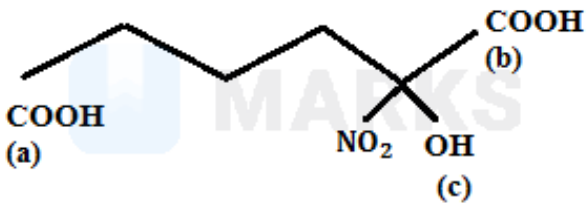
(B) 13.33%

(C) 33.33%

(D) 26.66%

Q.11

If 1 equivalent NaOH reacts with the given compound:



Choose the correct option:

(A) No reaction occurs.

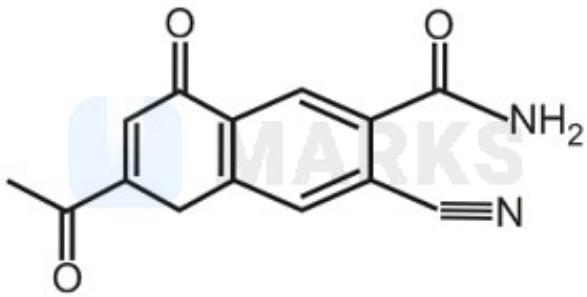
(B) $NaOH$ reacts with $-COOH$ (b)

(C) $NaOH$ reacts with $-COOH$ (a)

(D) $NaOH$ reacts with - OH

Q.12

Find out the double bond equivalent (DBE) value of the given following compound: < br /> < br /> < u> < strong>



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(A) 12.00

(B) 11.00

(C) 13.00

(D) 10.00

Q.13

The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of :

(A) $\mathrm{Fe}\left[\mathrm{Fe(CN)_6}\right]_3$

(B) $\mathrm{Na_3}\left[\mathrm{Fe(CN)_6}\right]$

(C) $\mathrm{Fe(CN)_3}$

(D) $\mathrm{Fe_4}\left(\mathrm{Fe(CN)_6}\right]_3$

Q.14

The purification of naphthalene can be done by

(A) filtration

(B) sublimation

(C) crystallization

(D) evaporation

Q.15

Two substances having slightly different solubility can be separated by

(A) distillation

(B) fractional distillation

(C) crystallization

(D) fractional crystallization

Q.16

Which of the following statements is true regarding the fractional distillation of a mixture of two liquids?

(A) The distillate collected is the liquid of lower boiling point

(B) The distillate collected is the liquid of higher boiling point

(C) The liquid left behind the flask is that of lower boiling point

(D) The compositions of distillate and the liquid left behind the flask do not depend upon the length of fractionating column

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Q.17

A compound which decomposes before its boiling point can be purified by the method of

(A) simple distillation

(B) fractional distillation

(C) distillation under reduce pressure

(D) steam distillation

Q.18

Phenol can be purified by the method of

(A) distillation

(B) steam distillation

(C) crystallization

(D) fractional distillation

Q.19

Essential oils can be isolated by

(A) crystallization

(B) steam distillation

(C) sublimation

(D) distillation

Q.20

Chromatography is based on the phenomenon of

(A) adsorption

(B) absorption

(C) solubility

(D) chemisorption

Answers & Solutions

Q.1 Answer:

Fractional distillation

Solution:

Fractional distillation

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Q.2 Answer:

catalytic agent

Solution:

Kjeldahl's **method depends upon the fact that most of the organic compounds containing nitrogen are quantitatively decomposed to give** $(NH_4)_2 SO_4$ **when heated strongly with conc.** H_2SO_4**. In this method** $CuSO_4$ **acts as catalytic agent.**

Q.3 Answer:

convert the covalent compound into a mixture of ionic compounds.

Solution:

To convert covalent compounds into ionic compounds such as $NaCN$, $Na_2 S$, NaX**, etc.**

Q.4 Answer:

sodium bicarbonate

Solution:

Carboxylic acids dissolve in sodium bicarbonate, while phenol does not.

Q.5 Answer:

Beilstein's Test

Solution:

Beilstein's test: Organic compounds containing halogens when heated over Cu **wire loop give blue or green colour flame due to formation of volatile copper halides.**

Q.6 Answer:

$Na_2 S$ and $NaCN$ are decomposed by HNO_3

Solution:

$Na_2 S$ **and** $NaCN$**, formed during fusion with metallic sodium, must be removed before adding** $AgNO_3$**, otherwise black ppt. due to** $Na_2 S$ **or white precipitate due to** $AgCN$ **will be formed and**

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thus white precipitate of $AgCl$ will not be identified easily.

$$Na_2S + 2AgNO_3 \longrightarrow 2NaNO_3 + \underset{\text{Black}}{Ag_2S\downarrow}$$

$$NaCN + AgNO_3 \longrightarrow NaNO_3 + \underset{\text{White}}{AgCN\downarrow}$$

$$NaCl + AgNO_3 \longrightarrow NaNO_3 + \underset{\text{white}}{AgCl\downarrow}$$

$$Na_2S + 2HNO_3 \xrightarrow{\text{boil}} 2NaNO_3 + H_2S\uparrow$$

$$NaCN + HNO_3 \xrightarrow{\text{boil}} NaNO_3 + HCN\uparrow$$

Q.7 Answer:

Sublimation

Solution:

Among the given compounds naphthelene is volatile but benzoic acid is non-volatile (it forms a dimer). So, the best method for their separation is sublimation,

Q.8 Answer:

steam distillation

Solution:

Aniline is miscible in steam and has b.p. lower than water, hence it is collected as distillate along with steam.

Q.9 Answer:

HCHO

Solution:

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% ratio of C : H is 6 : 1 and C : O is 3 : 4 or 6 : 8 $\therefore$ % ratio of C : H : O is 6 : 1 : 8 (Total $= 15$)

% of C $= \frac{6}{15} \times 100 = 40$

$40/12 = 3.33 = 1$

% of H $= \frac{1}{15} \times 100 = 6.6$

$6.6/1 = 6.6 = 2$

% of O $= \frac{8}{15} \times 100 = 53.3$

$53.3/16 = 3.3 = 1$

simple ratio: CH_2O

$\therefore$ **The compound is** HCHO

Q.10 Answer:

13.33%

Solution:

Percentage of $\vec{F}$

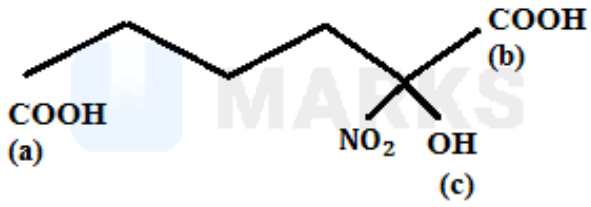
$= \frac{62}{222} \times \frac{\text{wt.of } Mg_2P_2O_7}{\text{wt.of compound}} \times 100$

$= \frac{62}{222} \times \frac{1.332}{2.79} \times 100 = 13.33\%$

Q.11 Answer:

NaOH reacts with -COOH (b)

Solution:



NaOH **is base so it will react with acid.**

H **which is attached with** -COOH **is more acidic due to the stability of conjugate base (equivalent resonating structure). So, b and c is more basic. And in a and b, b is more acidic because it's conjugate base is stabilised due to** $-I$ **effect of** $-NO_2$

Q.12 Answer:

11.00

Solution:

The double bond equivalent (DBE) value of the given compound is 11

Q.13 Answer:

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$Fe_4\,(Fe(CN)_6]_3$

Solution:

Reaction takes place in Lassaigne's test are as:

$Na + C + N \longrightarrow NaCN$

$6NaCN + FeSO_4 \longrightarrow Na_4\,[Fe(CN)_6] + Na_2\,SO_4$

$Na_4\,[Fe(CN)_6] + Fe^{+}3 \longrightarrow Fe_4\,(Fe(CN)_6]_3$

So, $Fe_4\,(Fe(CN)_6]_3$ is responsible for prussian blue colour obtained during the test of Nitrogen and Lassaigne test. O-nitrophenal is more volatile than pnitrophenal. There is intramolecular H - bonding in $O-$ nitrophenal while in $p-$ nitrophenal there is intermolecular hydrogen bonding which cause association of molecule. Intermolecular H - bondingincrease, boiling point increases while volatility reduces.

Q.14 Answer:

sublimation

Solution:

sublimation

Q.15 Answer:

fractional crystallization

Solution:

fractional crystallization

Q.16 Answer:

The distillate collected is the liquid of lower boiling point

Solution:

The distillate collected is the liquid of lower boiling point

Q.17 Answer:

distillation under reduce pressure

Solution:

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Q.18 Answer:

steam distillation

Solution:

steam distillation

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Q.19 Answer:

steam distillation

Solution:

steam distillation

Q.20 Answer:

adsorption

Solution:

adsorption

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